A Toeplitz positivity test on the angular distribution of zeta zeros, with an independent reproduction of the Connes–Consani–Moscovici spectral realization

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Abstract

The Keiper disk coordinate $z_\rho = 1 - 1/\rho$ places the nontrivial zeros of ζ on the unit circle exactly when the Riemann Hypothesis holds. We equip the resulting angular measure with its trigonometric moments and use finite Toeplitz positivity as a numerical detector of hypothetical off-critical zeros. Measured on the first 300 zeros with injected off-line quadruples $\rho = \frac{1}{2} + \delta + i\gamma$, the detector breaks positive semidefiniteness at matrix order N^* satisfying $N^* \log(1/r) \in [0.085,\ 0.34]$ across all tested (γ, δ), where $r = |z_\rho|$; this is five to ten times below the scalar sensitivity threshold $n \gtrsim \gamma^2/\delta$ of Voros for the Keiper–Li coefficients, and far below the exponential scale $1/\log(1/r)$. The minimal eigenvalue of the deviation-only Toeplitz matrix obeys the empirical law $\lambda_{\min} \approx -0.2\, N^3 \log^2(1/r)$. An on-circle control injection never triggers detection. We relate the mechanism to the Carathéodory–Fejér rigidity of nearly singular positive Toeplitz forms, the same structure used constructively in the spectral realization of Connes, Consani and Moscovici, and we report a full independent reproduction of their central computation at $(\lambda, N) = (3, 120)$: all twenty published eigenvalue–zero discrepancies are reproduced to every stated figure, and the two hypotheses of their main theorem, simplicity and evenness of the minimal Weil eigenvector, are verified numerically at this parameter point, with $\varepsilon_N = 2.954 \times 10^{-38}$ and spectral gap ratio 1.4×10^{-7}.

1 Introduction

Write $\rho = \beta + i\gamma$ for the nontrivial zeros of the Riemann zeta function and

$$z_\rho \;=\; 1 - \frac{1}{\rho}\,, \tag{1}$$

the conformal coordinate introduced by Keiper [1], which maps the critical line $\operatorname{Re} s = \frac{1}{2}$ to the unit circle. The Riemann Hypothesis is the statement $|z_\rho| = 1$ for every ρ, and the Keiper–Li coefficients

$$\lambda_n \;=\; \sum_\rho \Big[\, 1 - \Big(1 - \frac{1}{\rho}\Big)^n \,\Big] \tag{2}$$

are nonnegative for all $n \geq 1$ if and only if RH holds [2]. Bombieri and Lagarias [3] showed that this criterion is an instance of a general positivity principle for multisets and connected it to Weil's quadratic functional. Voros [4, 5, 6] determined the sharp large-n dichotomy: under RH, $\lambda_n \sim \frac{n}{2}(\log n + \gamma_E - 1 - \log 2\pi)$, while a zero at $\frac{1}{2} + t + iT$ off the line contributes an exponentially growing oscillation that becomes visible against the smooth part only for $n \gtrsim T^2/|t|$, an uncertainty-principle barrier for the scalar sequence.

This note treats the zeros' angular positions on the Keiper circle as a measure and interrogates that measure with the classical trigonometric moment problem. Three results are reported, all numerical and all obtained with two independent methods or explicit controls.

First, finite Toeplitz positivity of the empirical moments detects injected off-critical zeros at matrix orders well below the scalar barrier. For an injected quadruple at height γ and displacement δ, detection occurs at order N^* with $N^* \log(1/r) \in [0.085, 0.34]$ in every tested configuration (Table 2), where $\log(1/r) = \frac{1}{2}\log\big(1+8\delta/((2\delta-1)^2+4\gamma^2)\big)$ is the radial displacement in the disk coordinate. The scalar barrier corresponds to $n\log(1/r) \gtrsim 1$; the matrix test operates an order of magnitude below it because an $N \times N$ Toeplitz form correlates all moments up to order N simultaneously.

Second, the minimal eigenvalue of the Toeplitz matrix built from the deviation moments alone grows cubically: $\lambda_{\min} \approx -0.2\, N^3 \log^2(1/r)$ over the measured range (Section 4.3). The quadratic factor is the small-n expansion of the off-circle deviation $2\cosh(n\log r) - 2$; the cubic amplification in N is a property of the Toeplitz quadratic form that we state as a conjecture and pose as a problem in the perturbation theory of Toeplitz matrices.

Third, the detection mechanism is the contrapositive of a rigidity phenomenon that has recently been used constructively. The Carathéodory–Fejér structure theorem [12] forces the kernel polynomial of a singular positive Toeplitz matrix to have all its roots on the unit circle; the extension of this principle by Connes and van Suijlekom [16] underlies the spectral realization of low zeta zeros by Connes, Consani and Moscovici [17], where a nearly null vector of the truncated Weil quadratic form generates, with striking accuracy, the actual zeros. Section 5 reports a complete independent reproduction of the central computation of [17] at $(\lambda, N) = (3, 120)$, including numerical verification of the two standing hypotheses of their main theorem at this parameter point. The reproduction also isolates, through a deliberate perturbation, which part of their construction guarantees reality of the approximating spectrum and which part carries the arithmetic accuracy.

The detector is not proposed as a method for verifying RH in any range; direct methods locate zeros far more efficiently. Its interest is resolution-theoretic: it quantifies how much more a positivity criterion sees when it is read as a matrix condition rather than a scalar sequence, on the coordinate where the criterion is native.

2 The angular form of the Keiper–Li framework

All statements in this section are elementary restatements of (1) and (2) and are recorded for use in the sequel. With $m(x) = (x-1)/(x+1)$ one has the identity

$$z_\rho \;=\; m(2\rho - 1), \tag{3}$$

so the Keiper coordinate is the Möbius compression of the zero centered at the critical line. For an on-line zero $\rho = \frac{1}{2} + i\gamma$, $\gamma > 0$, the image lies on the unit circle at angle

$$\theta_\gamma \;=\; \pi - 2\arctan(2\gamma) \in (0, \pi), \tag{4}$$

and the conjugate zero contributes the angle $-\theta_\gamma$.

Proposition 2.1. *If all zeros with $|\gamma| \le \Gamma$ lie on the critical line, their contribution to λ_n is*

$$\lambda_n^{(\Gamma)} = \sum_{0<\gamma\le\Gamma} 4\sin^2\!\Big(\frac{n\,\theta_\gamma}{2}\Big).$$

Proof. For a conjugate pair on the circle, $1 - z^n + 1 - \bar{z}^n = 2 - 2\cos n\theta = 4\sin^2(n\theta/2)$, and (4) gives the angle. □

Proposition 2.2. *Let $w_\rho = \frac{1}{2}\,\mathrm{Log}(2\rho - 1)$ with the principal branch. Then ρ lies on the critical line if and only if $\operatorname{Im} w_\rho = \pm\pi/4$.*

Proof. $\operatorname{Re}(2\rho - 1) = 0$ if and only if $\mathrm{Log}(2\rho-1) \in \log|2\rho - 1| + i\{\pm\pi/2\}$. □

Under artanh, which inverts m, the coordinate w_ρ is the rapidity of the velocity z_ρ; RH states that all zero rapidities lie on the two horizontal lines $\operatorname{Im} w = \pm\pi/4$. Both propositions were confirmed numerically on the first 300 zeros: the identity (3) holds exactly in exact arithmetic, $\big||z_\rho| - 1\big| \le 1.3\times10^{-26}$ at working precision 25 digits, the angle formula (4) agrees with $\arg z_\rho$ to 4.4×10^{-26}, and $\operatorname{Im} w_\rho - \pi/4$ vanishes identically.

3 Two computations of λ_n

The experiments below require accurate λ_n for moderate n together with the zero angles. Two independent computations were run and compared.

Method A extracts λ_n as Taylor coefficients of $\frac{d}{dz}\log\xi\big(1/(1-z)\big)$ by discretized Cauchy integrals on the circle $|z| = 0.75$ with 256 nodes at 40-digit working precision, using $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$. No zeros enter this computation. The radius was chosen after observing, at radius 0.9, an aliasing contamination of size $\lambda_{n+256}\,r^{256} \approx 10^{-9}$; at $r = 0.75$ the extracted λ_1 agrees with the closed form $1 + \gamma_E/2 - \frac{1}{2}\log 4\pi$ to all fifteen printed digits. At fixed radius the extraction shares the well-known cancellation wall of this subject: rigorous large-n computation requires on the order of n bits of precision [10], and the present method is used only for $n \le 48$.

Method B evaluates Proposition 2.1 over the first 300 zeros and corrects the tail by the integral of the pair contribution $2\operatorname{Re}\big[1 - (1-1/\rho)^n\big]$ against the smoothed density $\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$.

n	Method A (zero free)	Method B (zeros + tail)	A − B
1	0.023095708966121	0.023099086453	-3.4×10^{-6}
2	0.092345735228047	0.092359245164	-1.4×10^{-5}
5	0.575542714461177	0.575627151058	-8.4×10^{-5}
13	3.81724005784795	3.81781082592	-5.7×10^{-4}
21	9.61796072314292	9.61945000892	-1.5×10^{-3}
48	40.8652694603672	40.8730461068	-7.8×10^{-3}

Table 1: Keiper–Li coefficients by the two methods. The difference carries a single sign and grows smoothly with n; it is the deterministic bias of replacing the discrete zero sum beyond height $\gamma_{300} \approx 541.8$ by the smoothed density, and it bounds the truncation systematics relevant to Section 4.

4 The Toeplitz detector

Let $\mu_K = \sum_{k\le K}(\delta_{\theta_k} + \delta_{-\theta_k})$ be the angular measure of the first K zero pairs and

$$c_n \;=\; \int e^{-in\theta}\, d\mu_K \;=\; 2\sum_{k\le K}\cos\bigl(n\theta_k\bigr) \tag{5}$$

its trigonometric moments. By the Herglotz–Toeplitz theorem [11], a two-sided sequence is the moment sequence of a positive measure on the circle if and only if every finite Toeplitz matrix $T_N = (c_{i-j})_{i,j=0}^{N-1}$ is positive semidefinite. Points on the circle therefore keep every T_N positive semidefinite regardless of RH considerations, while a zero off the line, entering with its symmetric quadruple ρ, $\bar\rho$, $1-\rho$, $1-\bar\rho$ and hence with velocities $z, \bar z, 1/z, 1/\bar z$ at radius $r \neq 1$, contributes

$$d_n \;=\; 2\cos(n\theta)\bigl(r^{\,n} + r^{-n}\bigr), \tag{6}$$

which is unbounded and must eventually break positivity.

The detector is: fix $K = 300$, inject one quadruple at $\rho = \frac{1}{2} + \delta + i\gamma$, and record the least N^* at which the smallest eigenvalue of T_N falls below $-10^{-6}c_0$.

4.1 Controls

Injecting a unit-circle atom at the same angle as the off-line test point, with the same total weight, never triggers detection: over matrix sizes up to 3000 the smallest eigenvalue stays above -1.9×10^{-10}, four orders of magnitude inside the tolerance. Detection therefore responds to off-circle displacement and not to the presence of an added atom. The baseline matrices are themselves numerically near-singular at all tested sizes, with $\lambda_{\min}$ of order 10^{-10} against $c_0 = 600$: the angles (4) accumulate at 0 like $1/\gamma$, and clustered atoms make the trigonometric moment problem nearly degenerate. This near-degeneracy is not an obstacle but the operative mechanism; see Section 5.

The threshold dependence is weak: for $(\gamma,\delta) = (10, 0.2)$, sweeping the tolerance from 10^{-3} to 10^{-8} moves N^* only from 208 to 82, an exponent of 0.081 per decade. Detection is governed by the geometry of the perturbed form against the near-null cluster space, not by threshold crossing of a growing scalar.

4.2 Resolution

The comparison in Table 2 is against a heuristic threshold supported by saddle-point analysis rather than a theorem with constants, and it should be read at that strength. The mechanism of the gain is nevertheless transparent: a single coefficient λ_n cannot resolve a radial displacement until $n\,|\log r| \gtrsim 1$, which is the uncertainty principle in the conjugate pair $(n, \log|z|)$ made explicit in [5]; the quadratic form on all moments up to N is subject to no such bound and instead feels the deviation (6) at its quadratic onset $d_n - 4\cos(n\theta) \approx 2\cos(n\theta)\, n^2 \log^2 r$.

4.3 The eigenvalue law

Isolating the deviation, the Toeplitz matrices built from $d_n - 4\cos(n\theta)$ alone have smallest eigenvalues, at $(\gamma,\delta) = (10, 0.2)$,

$$-0.119,\; -0.790,\; -6.10,\; -46.3,\; -401 \quad \text{at} \quad N = 50, 100, 200, 400, 800,$$

γ	δ	$\log(1/r)$	N^*	$N^* \log(1/r)$
10	0.40	3.98×10^{-3}	83	0.331
10	0.30	2.99×10^{-3}	91	0.272
10	0.20	1.99×10^{-3}	111	0.221
10	0.10	9.97×10^{-4}	140	0.140
10	0.05	4.99×10^{-4}	171	0.085
5	0.20	7.91×10^{-3}	27	0.214
20	0.20	5.00×10^{-4}	673	0.336
14.13	0.25	1.25×10^{-3}	154	0.193

Table 2: Detection order for injected off-line quadruples. In every case $N^* \log(1/r) < 0.34$: detection occurs while the exponential factor r^{-n} has grown by less than 40 percent. The scalar visibility threshold of [4, 5] is $n \log(1/r) \gtrsim 1$, equivalently $n \gtrsim \gamma^2/\delta$; for the row $(\gamma, \delta) = (10, 0.2)$ that threshold is $n \approx 500$ against the measured $N^* = 111$, and the same factor of five to ten separates the two across the table.

a factor close to 8 per doubling.

Conjecture 4.1. *For the deviation symbol of a single off-circle quadruple at radius r and angle θ, as $N \to \infty$ with $N \log(1/r) \to 0$,*

$$\lambda_{\min}\big(T_N[d - 4\cos(\cdot\,\theta)]\big) \;=\; -\,c(\theta)\, N^3 \log^2(1/r)\,\big(1 + o(1)\big),$$

with $c(\theta)$ of order 0.2 at the measured configuration.

Question 4.2. *Derive Conjecture 4.1, including the constant, from the asymptotic spectral theory of Toeplitz matrices with polynomially growing symbol coefficients [13].*

5 Relation to the Carathéodory–Fejér mechanism, and a reproduction of the spectral realization of Connes, Consani and Moscovici

The structure theorem of Carathéodory and Fejér [12] states that a positive semidefinite Toeplitz matrix with nontrivial kernel has a kernel polynomial whose roots all lie on the unit circle. Read contrapositively, a nearly singular positive Toeplitz form has no room for off-circle mass: any such mass must break positivity against the near-null space essentially at once, rather than after exponential amplification. This is the phenomenon measured in Section 4, where the clustered zero angles supply the near-degeneracy and the injected quadruple supplies the off-circle mass.

The same structure is used in the opposite direction by Connes, Consani and Moscovici [17], building on [14, 15] and on the extension of the Carathéodory–Fejér principle in [16]. There, the input is arithmetic rather than spectral: the Weil quadratic form restricted to test functions on $[\lambda^{-1}, \lambda]$, involving only the prime powers up to λ^2, is truncated to $2N{+}1$ Fourier modes; the eigenvector ξ of its smallest eigenvalue ε_N, made an exact kernel vector by subtracting ε_N, turns a rank-one perturbation of the scaling operator into an operator that is self-adjoint for the inner product defined by the form. Its spectrum, which is the real zero set of the Fourier transform $\widehat{\xi}$, is real by construction, and it approximates the actual ordinates of the low zeta zeros to remarkable accuracy.

We reproduced the computation of [17] at $(\lambda, N) = (3, 120)$ independently and from scratch: all closed-form matrix-element formulas were re-derived and verified against direct quadrature before use, the matrix was assembled through its structure $\tau_{ij} = (b_i - b_j)/(i - j)$ for $i \neq j$, the problem was reduced by parity, and the near-kernel eigenpair was extracted by inverse iteration with a 200-digit LU factorization. The constant $c(L) = \int_0^L (e^{x/2} - 1)/(e^x - e^{-x})\,dx = 0.575219384657911$ at $L = 2\log 3$ enters every diagonal entry through $2c(L)$.

k	this work	[17]	k	this work	[17]
1	1.58×10^{-34}	1.6×10^{-34}	11	7.28×10^{-17}	7.3×10^{-17}
2	2.06×10^{-31}	2.1×10^{-31}	12	2.21×10^{-15}	2.2×10^{-15}
3	1.46×10^{-29}	1.5×10^{-29}	13	9.75×10^{-14}	9.7×10^{-14}
4	8.29×10^{-27}	8.3×10^{-27}	14	2.70×10^{-13}	2.7×10^{-13}
5	1.28×10^{-25}	1.3×10^{-25}	15	6.34×10^{-12}	6.3×10^{-12}
6	1.17×10^{-23}	1.2×10^{-23}	16	5.63×10^{-11}	5.6×10^{-11}
7	7.53×10^{-22}	7.5×10^{-22}	17	2.93×10^{-10}	2.9×10^{-10}
8	6.57×10^{-21}	6.6×10^{-21}	18	1.22×10^{-9}	1.2×10^{-9}
9	1.19×10^{-18}	1.2×10^{-18}	19	5.58×10^{-8}	5.6×10^{-8}
10	8.79×10^{-18}	8.8×10^{-18}	20	2.42×10^{-7}	2.4×10^{-7}

Table 3: Absolute differences between the ordinates γ_k of the first twenty zeta zeros and the spectrum of the perturbed scaling operator at $(\lambda, N) = (3, 120)$: the present reproduction against the values reported in Figure 1 of [17]. Agreement holds at every stated figure.

Two hypotheses of the main theorem of [17] are assumed there for general parameters: that ε_N is simple and that ξ is even. Both are facts at this parameter point:

$$\varepsilon_N \;=\; 2.954058093 \times 10^{-38}, \qquad \frac{\varepsilon_N}{\lambda_2^{\text{even}}} \;=\; 1.38 \times 10^{-7}, \qquad \lambda_{\min}^{\text{odd}} \;=\; 1.13 \times 10^{-34},$$

so the smallest eigenvalue is simple with seven orders of separation inside the even sector and sits four orders below the entire odd sector. The size of ε_N is consistent with the superexponential decay displayed in Figure 4 of [17].

Remark 5.1. The reproduction separates the two guarantees of the construction. A bookkeeping error in the eigenvector coordinates, a factor $\sqrt{2}$ between the parity-reduced and the original basis, leaves every approximating eigenvalue exactly real while destroying all arithmetic accuracy, the discrepancies in Table 3 inflating from 10^{-34} to order one. Reality of the spectrum is structural and survives arbitrary corruption of ξ; proximity to the Riemann zeros lives entirely in the fine structure of the near-kernel vector. The detector of Section 4 and the realization of [17] are in this sense two readings of one rigidity: the near-null space of a positive form on a finite window is, at once, the most sensitive locus for off-line mass and the most information-dense encoding of the on-line zeros.

6 Scope and next steps

The truncation systematics of the detector are bounded by Table 1; the injected-quadruple experiments are synthetic, and the planned validation is the Davenport–Heilbronn function, which satisfies a functional equation while violating the analogue of RH and therefore supplies genuine off-line zeros, as used for the discretized Keiper–Li sequences in [6]. On the theoretical side the open items are Conjecture 4.1 and the extension of the detection-order analysis to a statement with constants, for which the natural benchmark is the Oesterlé bound quoted in [5], that verification of RH up to height T_0 forces $\lambda_n \geq 0$ for $n \leq T_0^2$. The nearest published relatives of the detector are the Fourier-positivity tests of Giraud and Peschanski [18], which watch the lowest eigenvalue of a Toeplitz hierarchy for a candidate positive-definite function, and the model-space norm identity of Suzuki [19], which realizes λ_n as a squared norm; the present test differs from the first in its object, the empirical angular measure of the zeros, and from the second in being a finite-order matrix condition rather than an identity.

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